

JAE HONG LEE

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EDUCATION

Boston University, Boston, MA

B.A., Computer Science / Minor in Visual Arts

Computing and Technical Honor Society / Upsilon Pi Epsilon (UPE)

Expected May 2025

3.89 GPA | Dean's List (x4)

Korea University, Seoul, Korea

Department of Computer Science and Engineering

Sep 2022 - Jun 2023

Study Abroad

PUBLICATIONS

Jongin Choi, Jaehong Lee, Daniel Oh, and Eung-Joo Lee, "Machine Learning-Based Hand Pose Generation Using a Haptic Controller." *Electronics* 13.10 (2024): 1970

PROFESSIONAL EXPERIENCE

Quantum Computing Research Intern

Jun 2024 – Aug 2024

Korea Electronics Technology Institute

Seungnam, Korea

- Contributed to research optimizing memory efficiency for quantum simulations by storing only measured qubits up to 45 qubits, and developed a quantum circuit simulator using Google's Tensor Network, qiskit, and statevector to perform Grover's algorithm.
- Acquired comprehensive knowledge in quantum computing through foundational studies and delivered two internal seminars on key concepts such as qubits, superposition, Bell's theorem, the EPR paradox, and quantum computing simulations.

Computer Vision Research Intern

Jun 2022 – Aug 2022

Korea Electronics Technology Institute

Seungnam, Korea

- Applied Google's Vision Transformer (ViT) for advanced breed classification in cats and dogs kaggle dataset, incorporating patch based Transformer and Multi-head Attention mechanisms, achieving a classification accuracy of over 95% on benchmark datasets.
- Re-engineered the ViT model using TensorFlow, developing an API and integrating it with a open CV camera module for real-time capture applications, enabling accurate breed classification for real-life pets, improving both speed and precision.

Software Engineer Intern

Jul 2020 – Nov 2020

Xenix Studio

Seoul, Korea

- Integrated Blockchain technology into an on-site restaurant payment application using smart contracts, ensuring only verified diners could enhance the authenticity of in-service scores, while deepening my expertise in cryptocurrency and decentralization.
- Developed a responsive web application for Gandago, enabling them to enhance their online presence and engage customers across multiple devices, using Figma, React.js, HTML, CSS, and Bootstrap.

RESEARCH PROJECTS

Text-to-Panorama Generation Undergraduate Research

April 2024 - Present

Research Assistant, Advisor: Aoming Liu

Boston, MA

- Investigating a cube-based approach to refine text-to-360-degree panorama generation by combining spot diffusion techniques with latent space noise projection, with the goal of creating depth and improving visual continuity.
- Focused on achieving seamless transitions and realism in panorama generation by completing a 3D camera tutorial, covering areas of camera calibration, object localization, and geometric distortion correction to improve accuracy and depth perception.

AI-driven Hand Pose estimation Undergraduate Research

Sep 2023 - May 2024

Research Assistant, Advisor: Eung-Joo Lee, Jong In Choi

- Implemented a custom-built haptic controller to generate natural hand postures, trained the XGBoost model with a collection of 3000-9000 training datasets, which improved handpose accuracy as dataset size increased.
- Designed a technique for generating hand poses and identified the need for time-series methods to reduce tremors in haptic control.

SOFTWARE ENGINEER PROJECTS

To-Do Calendar

Jan 2024 – May, 2024

React.js, Google Calender & Authentication API, Firebase, CSS, Figma

Boston, MA

- Integrated task management with Google Calendar to sync tasks and deadlines, improving productivity by providing a overview.
- Automatic meeting additions have been implemented for Google Meet and Zoom to ensure users won't miss a meeting and to visually link tasks to their respective deadlines.

TECHNICAL SKILLS

Programming: C, Python, Julia, R, Java, MATLAB, Kotlin, Verilog, MySQL, HTML, CSS, JavaScript, Shell

Frameworks and Libraries: React.js, TensorNetwork, Hugging Face, Tensorflow, Pytorch, Sklearn, OpenCV, Docker, Flask/Django

Developer and Design Tools: CUDA, AWS, Linux, Azure, Figma, Sketch, Zeplin, Adobe Tool, Latex, Microsoft Office

Concepts: Software Engineering, Artificial Intelligence, Quantum Computing, Machine Learning, Neural Networks, Computer Vision, Computer Network, Distributed System, Database, Human Computer Interaction, Algorithms, Data Structure, Graphic Design